

General Introduction Algorithms

Let's write a program

A program is a collection of statements written in a language the computer understands.

A computer executes program statements one after another **in sequence** until it reaches the end of the program unless some statement in the program alters the order of execution.

Let's write a program to accept two numbers from the user, multiply them together and display the result on the screen.

We'll write the program in COBOL.

What COBOL program statements will we need to do the job?



Program Statements

We need a statement to take in the first number and store it in a named memory location.

ACCEPT Num1.

We need a statement to take in the second number and store it in a named memory location.

ACCEPT Num2.

We need a statement to multiply the two numbers together and store the result in a named location.

MULTIPLY Num1 BY Num2 GIVING Result.

We need a statement to display the result on the screen.

DISPLAY "Result is = ", Result.



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

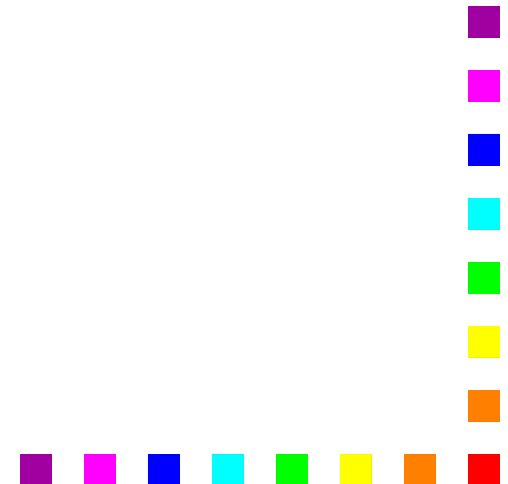
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 0.  
01  Num2          PIC 9   VALUE 0.  
01  Result        PIC 99  VALUE 0.
```

Calc-Result

DATA		
Num1	Num2	Result
0	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

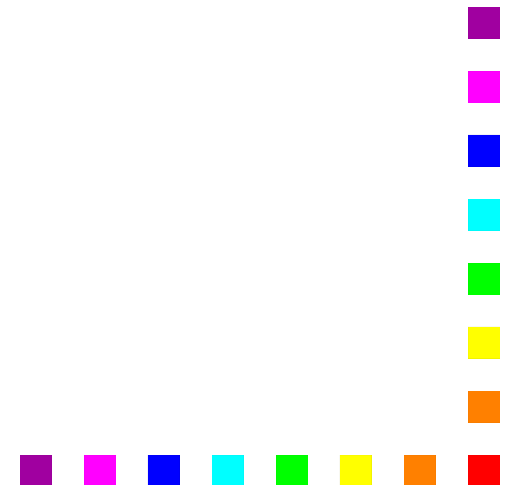
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 9.  
01  Num2          PIC 9   VALUE 0.  
01  Result        PIC 99  VALUE 00.
```

```
ACCEPT Num1
```

DATA		
Num1	Num2	Result
9	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

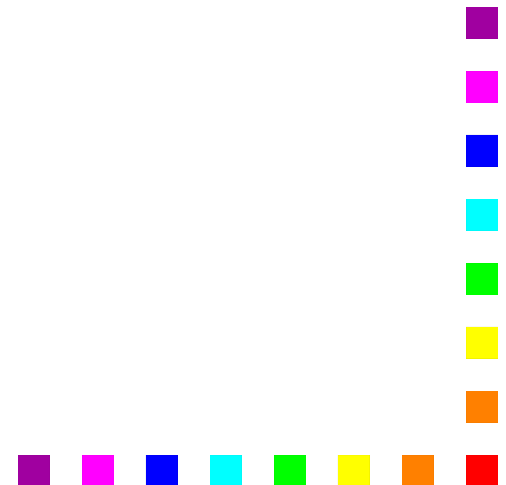
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE 9.  
01  Num2          PIC 9  VALUE 0.  
01  Result        PIC 99 VALUE 00.
```

DATA		
Num1	Num2	Result
9	0	00

```
MULTIPLY Num1 BY Num2 GIVING Result
```



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

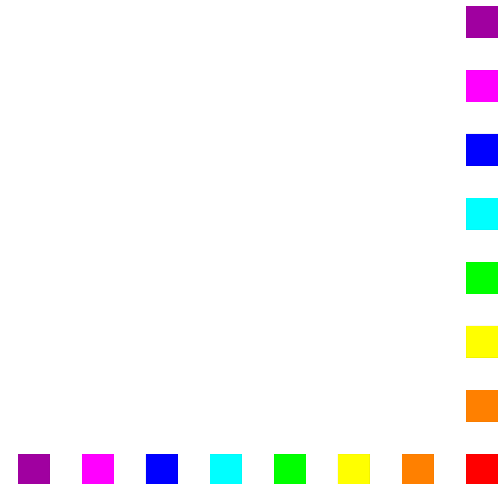
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 0.  
01  Num2          PIC 9   VALUE 5.  
01  Result        PIC 99  VALUE 0.
```

```
ACCEPT Num2
```

DATA		
Num1	Num2	Result
0	5	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentA.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

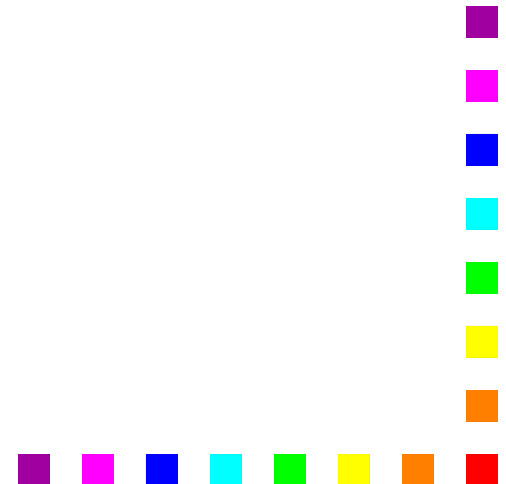
```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 9.  
01  Num2          PIC 9   VALUE 5.  
01  Result        PIC 99  VALUE 00.
```

```
DISPLAY "Result is = ", Result
```

DATA		
Num1	Num2	Result
9	5	00

**Oh No!!
We got the
wrong answer.
Let's try again.**




```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. FragmentB.  
AUTHOR. Michael Coughlan.
```

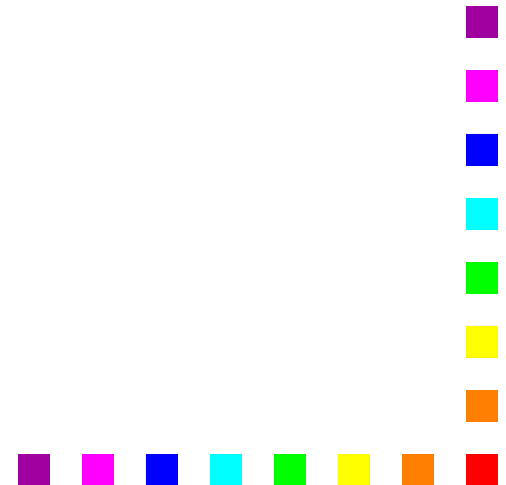
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE  
01  Num2          PIC 9  VALUE  
01  Result        PIC 99 VALUE
```

Calc-Result

DATA		
Num1	Num2	Result
0	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

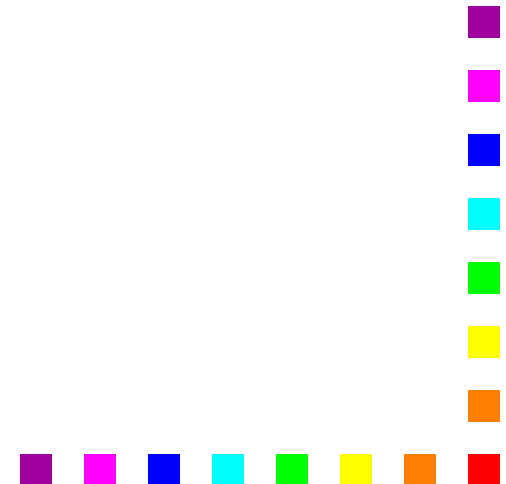
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 9.  
01  Num2          PIC 9   VALUE 0.  
01  Result        PIC 99  VALUE 00.
```

```
ACCEPT Num1.
```

DATA		
Num1	Num2	Result
9	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

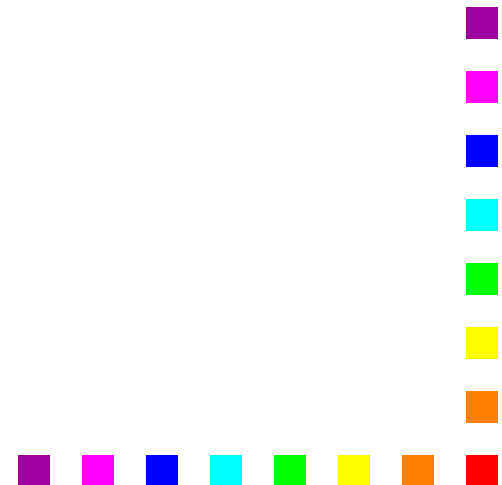
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 9.  
01  Num2          PIC 9   VALUE 5.  
01  Result        PIC 99  VALUE 00.
```

```
ACCEPT Num2
```

DATA		
Num1	Num2	Result
9	5	00



```
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IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

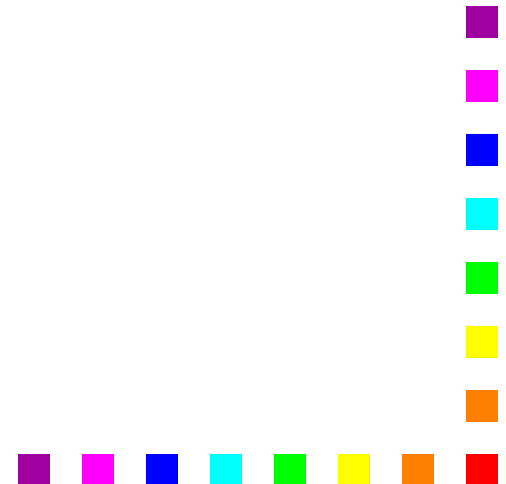
```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE 9.  
01  Num2          PIC 9  VALUE 5.  
01  Result        PIC 99 VALUE 00.
```

```
DISPLAY "Result is = ", Result
```

DATA		
Num1	Num2	Result
9	5	00

**Oh No!!
We got the wrong
answer again.
Let's have one
more try.**



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    FragmentB.  
AUTHOR.    Michael Coughlan.
```

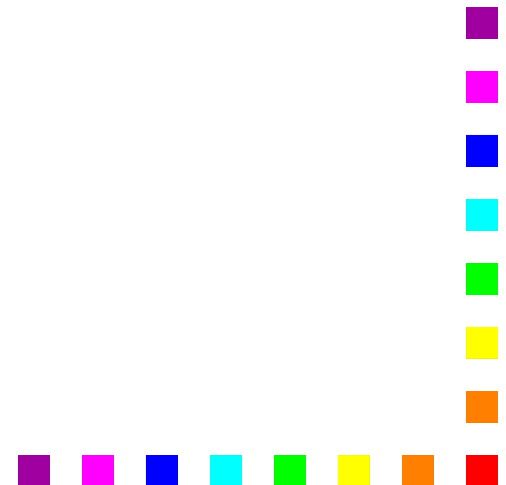
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 9.  
01  Num2          PIC 9   VALUE 5.  
01  Result        PIC 99  VALUE 0.
```

DATA		
Num1	Num2	Result
9	5	45

MULTIPLY Num1 BY Num2 GIVING Result



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program  
AUTHOR.    Michael Coughlan.
```

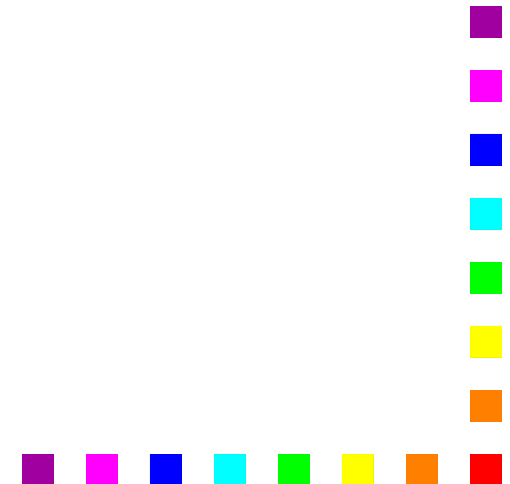
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE 0  
01  Num2          PIC 9  VALUE 0  
01  Result        PIC 99 VALUE 0
```

```
Calc-Result
```

DATA		
Num1	Num2	Result
0	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

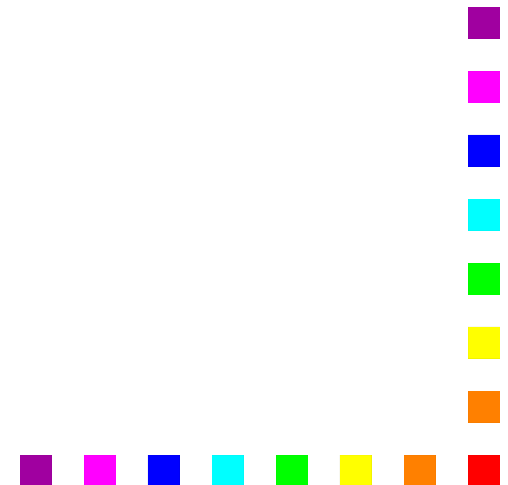
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 9.  
01  Num2          PIC 9   VALUE 0.  
01  Result        PIC 99  VALUE 00.
```

```
ACCEPT Num1
```

DATA		
Num1	Num2	Result
9	0	00



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

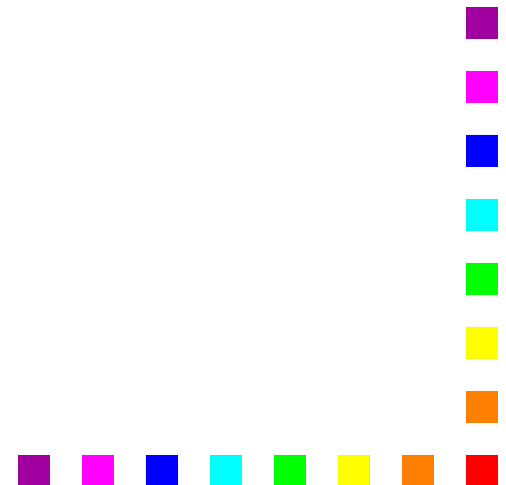
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE 9.  
01  Num2          PIC 9  VALUE 5.  
01  Result        PIC 99 VALUE 00.
```

```
ACCEPT Num2
```

DATA		
Num1	Num2	Result
9	5	00




```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

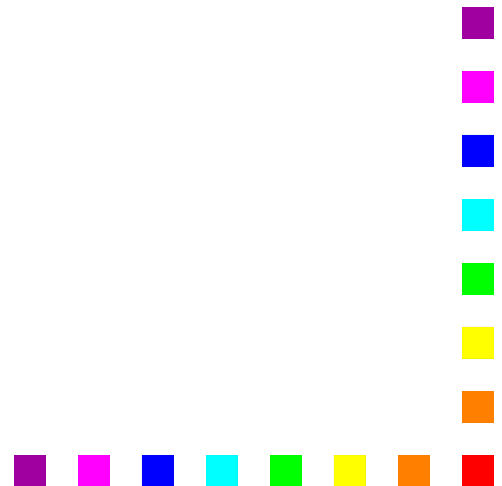
```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9  VALUE 9.  
01  Num2          PIC 9  VALUE 5.  
01  Result        PIC 99 VALUE 0.
```

DATA		
Num1	Num2	Result
9	5	45

MULTIPLY Num1 BY Num2 GIVING Result



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Sequence-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9   VALUE 0.  
01  Num2          PIC 9   VALUE 5.  
01  Result        PIC 99  VALUE 0.
```

```
DISPLAY "Result is = ", Result
```

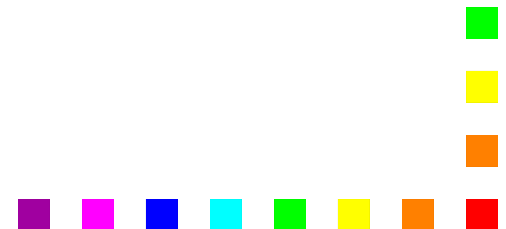
DATA

Num1	Num2	Result
------	------	--------

0	5	45
---	---	----

Right .. At last!!

It's not enough to know
what statements we
need. We must make
sure the computer
executes them in the
right order.

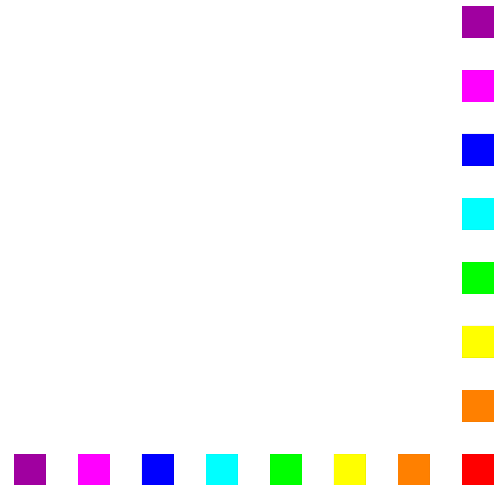


Programs are written using three main programming constructs.

Sequence.

Selection.

Iteration.

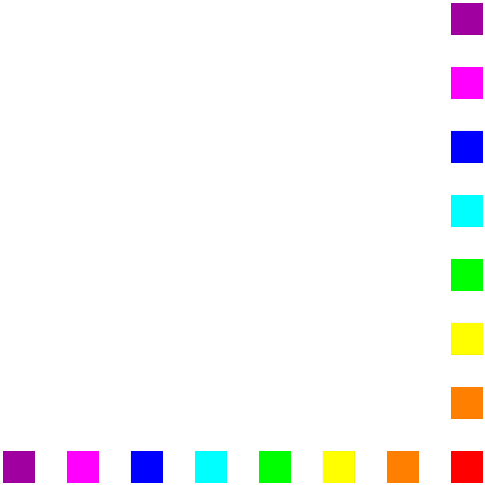


\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Selection-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

DATA			
Num1	Num2	Result	Operator
0	0	00	-

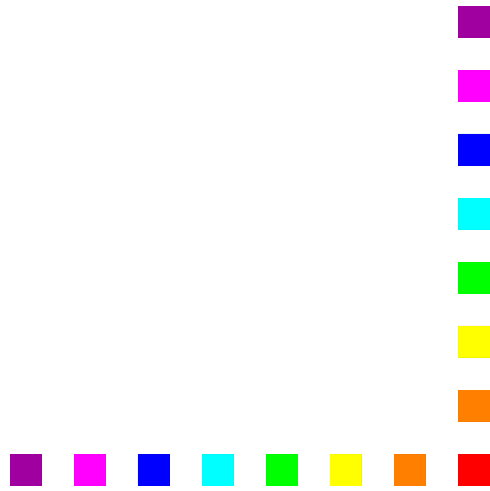


\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Selection-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator
ACCEPT Num1

DATA			
Num1	Num2	Result	Operator
7	0	00	-



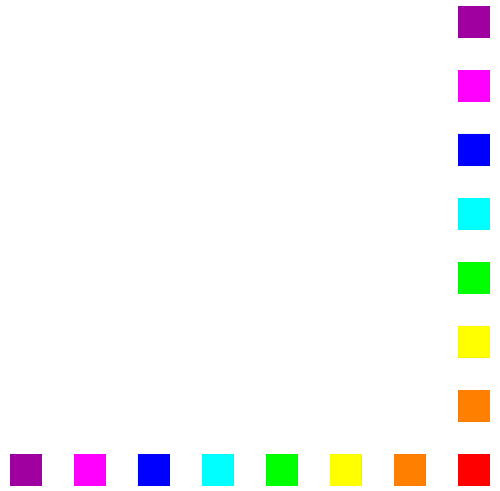
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Selection-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

ACCEPT Num2

DATA			
Num1	Num2	Result	Operator
7	3	00	-



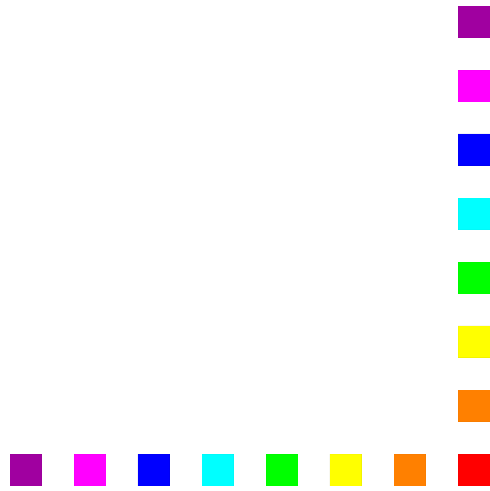
```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Selection-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01  Num1          PIC 9  VALUE  
01  Num2          PIC 9  VALUE  
01  Result        PIC 99 VALUE  
01  Operator      PIC X  VALUE
```

```
PROCEDURE DIVISION.  
Calculator.
```

ACCEPT Operator

DATA			
Num1	Num2	Result	Operator
7	3	00	+



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Selection-Program.  
AUTHOR. Michael Coughlan.
```

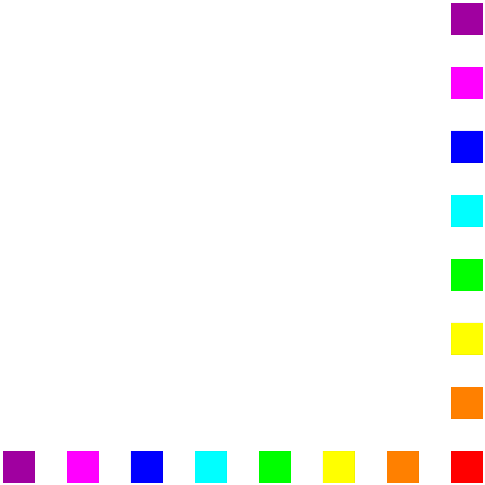
```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 Num1 PIC 9 VALUE  
01 Num2 PIC 9 VALUE  
01 Result PIC 99 VALUE  
01 Operator PIC X VALUE
```

```
PROCEDURE DIVISION.  
Calculator.
```

IF Operator = "+" THEN

END-IF

DATA			
Num1	Num2	Result	Operator
7	3	00	+



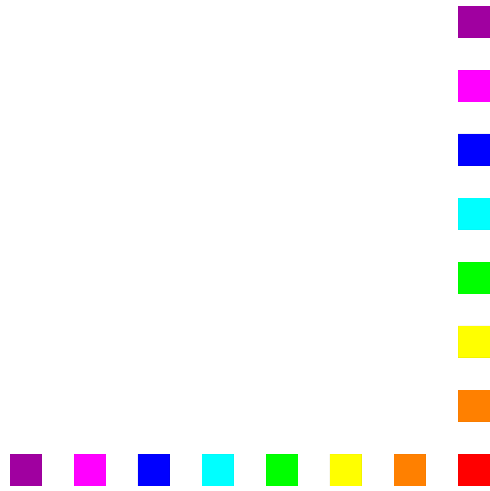
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Selection-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE 7
01 Num2 PIC 9 VALUE 3
01 Result PIC 99 VALUE 10
01 Operator PIC X VALUE +

PROCEDURE DIVISION.
Calculator.

ADD Num1, Num2 GIVING Result

DATA			
Num1	Num2	Result	Operator
7	3	10	+



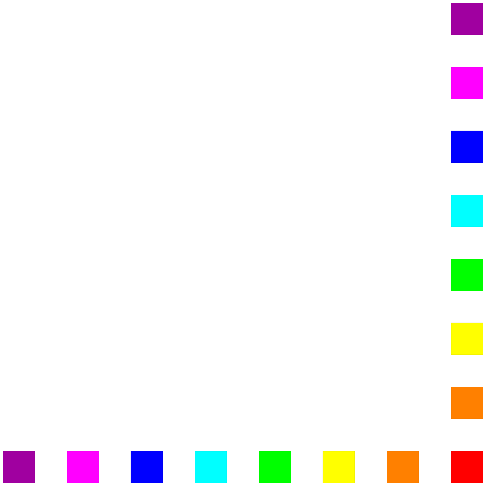
```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    Selection-Program.
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
WORKING-STORAGE SECTION.
01  Num1          PIC 9  VALUE
01  Num2          PIC 9  VALUE
01  Result        PIC 99 VALUE
01  Operator      PIC X  VALUE
```

```
PROCEDURE DIVISION.
Calculator.
```

```
IF Operator = "*" THEN
END-IF
```

DATA			
Num1	Num2	Result	Operator
7	3	10	+



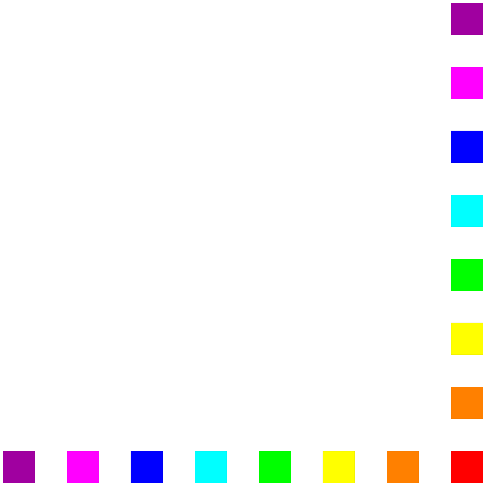
```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Selection-Program.  
AUTHOR. Michael Coughlan.
```

```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 Num1 PIC 9 VALUE 7  
01 Num2 PIC 9 VALUE 3  
01 Result PIC 99 VALUE 0  
01 Operator PIC X VALUE +
```

```
PROCEDURE DIVISION.  
Calculator.
```

```
DISPLAY "Result is = ", Result
```

DATA			
Num1	Num2	Result	Operator
7	3	10	+

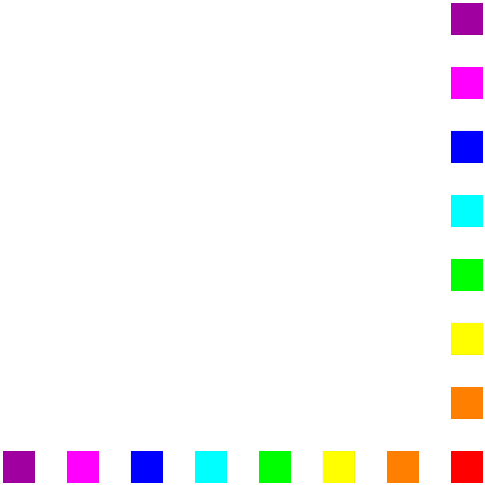


\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. **Iteration-Program**.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

DATA			
Num1	Num2	Result	Operator
0	0	00	—



\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

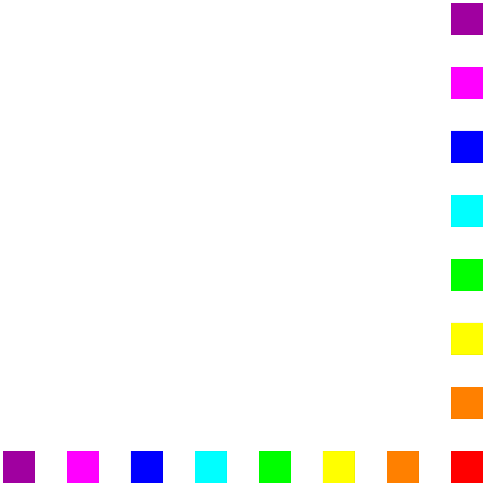
DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

PERFORM 5 TIMES

END-PERFORM

DATA			
Num1	Num2	Result	Operator
0	0	00	—



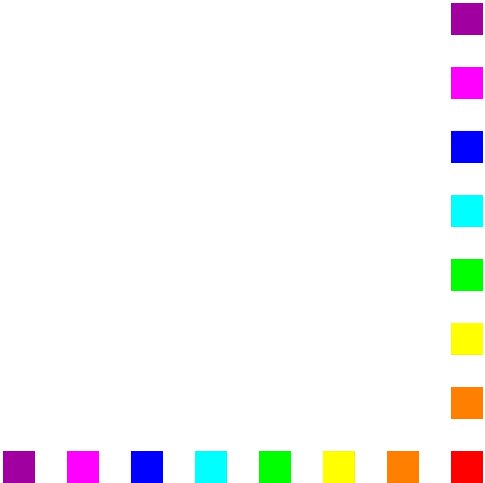
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

ACCEPT Num1

DATA			
Num1	Num2	Result	Operator
5	0	00	—



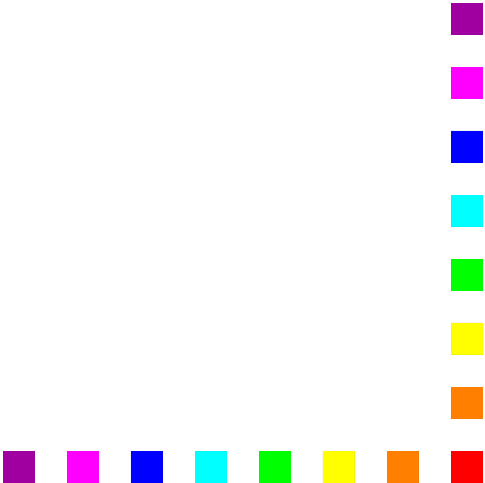
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

ACCEPT Num2

DATA			
Num1	Num2	Result	Operator
5	3	00	—



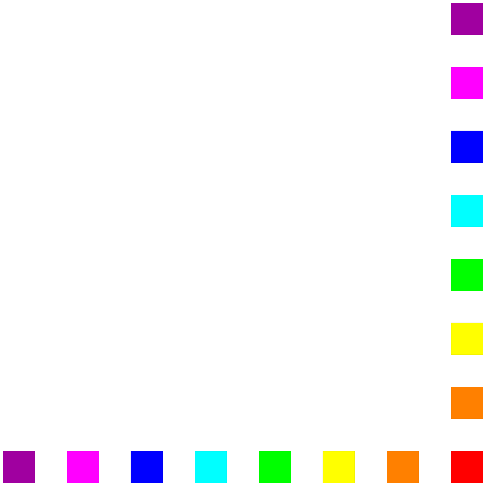
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

ACCEPT Operator

DATA			
Num1	Num2	Result	Operator
5	3	00	*




```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Iteration-Program.  
AUTHOR. Michael Coughlan.
```

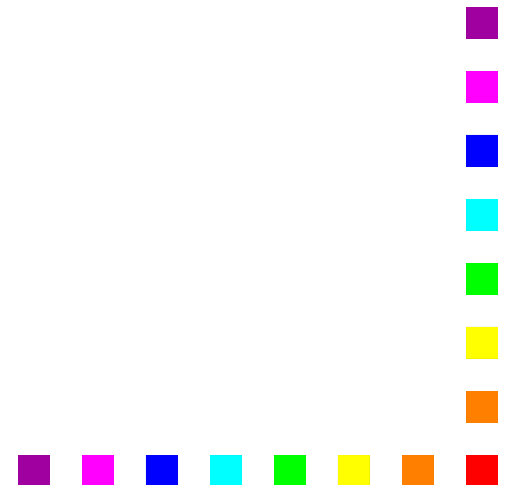
```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 Num1 PIC 9 VALUE  
01 Num2 PIC 9 VALUE  
01 Result PIC 99 VALUE  
01 Operator PIC X VALUE
```

```
PROCEDURE DIVISION.  
Calculator.
```

```
IF Operator = "+" THEN
```

```
END-IF
```

DATA			
Num1	Num2	Result	Operator
5	3	00	*



```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Iteration-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.
```

```
WORKING-STORAGE SECTION.
```

```
01  Num1          PIC 9    VALUE  
01  Num2          PIC 9    VALUE  
01  Result        PIC 99   VALUE  
01  Operator      PIC X    VALUE
```

```
PROCEDURE DIVISION.
```

```
Calculator.
```

```
    ACCEPT Num1.
```

```
    ACCEPT Num2.
```

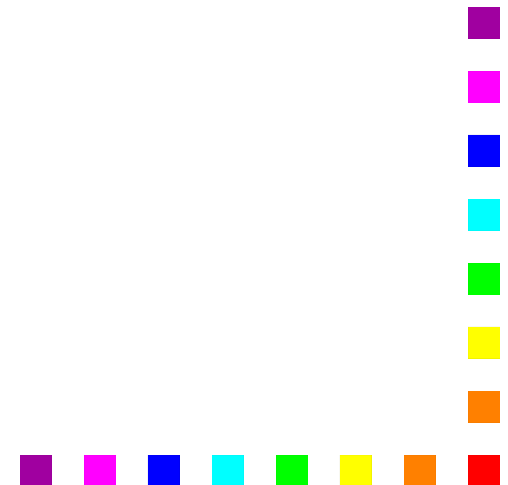
```
    ACCEPT Result.
```

```
    ACCEPT Operator.
```

```
    IF Operator = "*" THEN
```

```
        END-IF
```

DATA			
Num1	Num2	Result	Operator
5	3	00	*



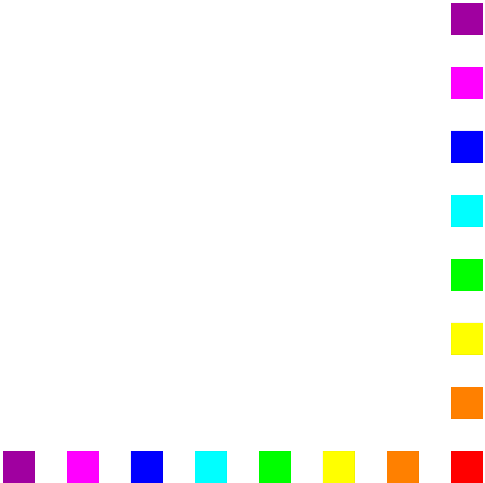
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

MULTIPLY Num1 BY Num2 GIVING Result

DATA			
Num1	Num2	Result	Operator
5	3	15	*



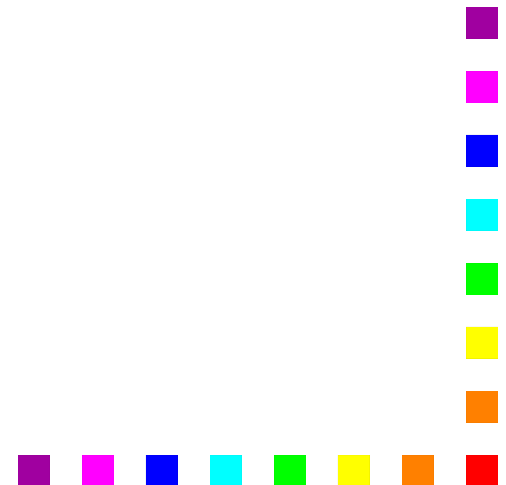
```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID.    Iteration-Program.  
AUTHOR.    Michael Coughlan.
```

```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01  Num1          PIC 9    VALUE  
01  Num2          PIC 9    VALUE  
01  Result        PIC 99   VALUE  
01  Operator      PIC X    VALUE
```

```
PROCEDURE DIVISION.  
Calculator.
```

```
DISPLAY "Result is = ", Result
```

DATA			
Num1	Num2	Result	Operator
5	3	15	*



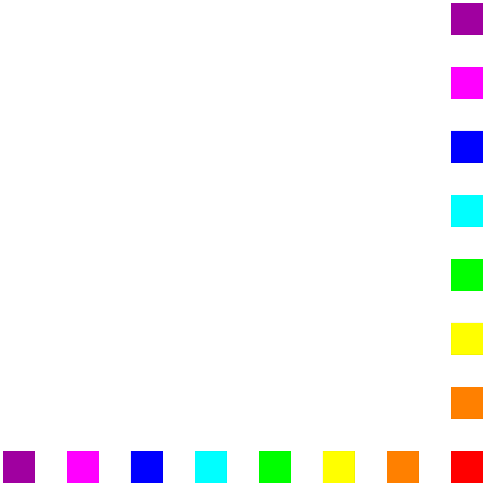
\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

ACCEPT Num1

DATA			
Num1	Num2	Result	Operator
9	3	15	*



\$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Iteration-Program.
AUTHOR. Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 Num1 PIC 9 VALUE
01 Num2 PIC 9 VALUE
01 Result PIC 99 VALUE
01 Operator PIC X VALUE

PROCEDURE DIVISION.
Calculator.

ACCEPT Num2

DATA			
Num1	Num2	Result	Operator
9	3	15	*

